



Data Exploration & Visualization - Business Case

Content Strategy Insights for Growth Across Markets

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Google Drive Link:

https://drive.google.com/file/d/1iUKAOpti9aTaOmo7IJNVZA3bMTAXrvni/view?usp=drive_link

1. Problem Statement & Basic Metrics

Business Problem Statement

Netflix wants to know what type of content it should make, whether Movies or TV Shows, which genres are the most popular, and how it can improve its content strategy in different countries. This analysis uses historical Netflix catalog data to find patterns in content type, genres, release years, countries, directors, and actors. The insights can help Netflix make better decisions about content production and expanding its content in different regions.

Metric	Value
Total titles	8,807
Total columns/attributes	12
Movies	6,131 (69.6%)
TV Shows	2,676 (30.4%)
Release year range	1925 - 2021
Titles added to platform (year range)	2008 - 2021
Duplicate rows	0

Observations:

- The dataset has 8,807 rows and 12 columns (attributes).
- No duplicate rows
- The dataset has missing values in several important columns like director, cast, and country.

2. Data Shape, Types, Missing Values & Statistical Summary

Shape and Data Types

- The dataset has 8,807 rows and 12 columns.
- Only column 'release_year' is numeric while others columns are of text/str data type.
- Columns 'date_added' is stored as text and is converted to datetime data type.

- The 'type' and 'rating' columns have only a few unique values, so they can be converted to the category data type.

Missing Value Detection

Column	Missing Count
director	2,634
cast	825
country	831
date_added	10
rating	4
duration	3

Observations: The 'director' column has the most missing values (~30%). The 'cast' and 'country' columns each have about ~9% missing values, while 'date_added', 'rating', and 'duration' have less than 1% missing values.

Statistical Summary (Numeric Attributes)

Stats	Value (release_year)
count	8,807
mean	2014.2
std dev	8.8
min	1925
25%	2013
median (50%)	2017
75%	2019
max	2021

Observation: The dataset includes titles released between 1925 and 2021, but most of them are recent. The median release year is 2017, and 75% of the titles were released in 2013 or later. This shows that Netflix mainly focuses on adding newer content rather than older movies and TV shows.

3. Non-Graphical Analysis

Value Counts - Key Categorical Fields

Field	Top Values
All Movies/TV Shows	Movie: 6131 TV Show: 2676
Top 5 ratings	TV-MA: 3207 (36.4%) TV-14: 2160 (24.5%) TV-PG: 863 (9.8%) R: 799 (9.1%) PG-13: 490 (5.6%)
Top 5 countries (unnested)	United States: 3689 (36.8%) India: 1046 (10.4%) United Kingdom: 804 (8%) Canada: 445 (4.4%) France: 393 (3.9%)
Top 5 genres (unnested)	International Movies: 2752 Dramas: 2427 Comedies: 1674 International TV Shows: 1351 Documentaries: 869
Top director (by title count)	Rajiv Chilaka: 22 titles
Top actor (by title count)	Anupam Kher: 43 titles

Unique Attribute Counts

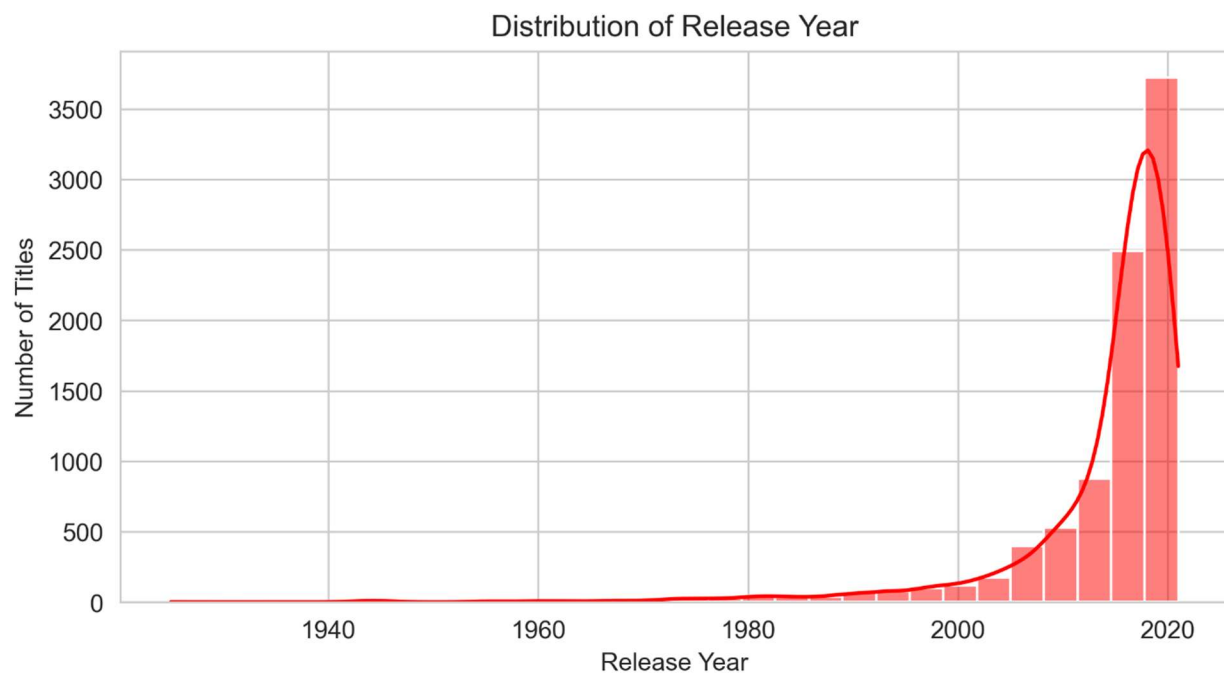
- Unique Countries: 127
- Unique Genres: 42
- Unique Cast Members (Actors): 36,439
- Unique Directors: 4993

4. Visual Analysis

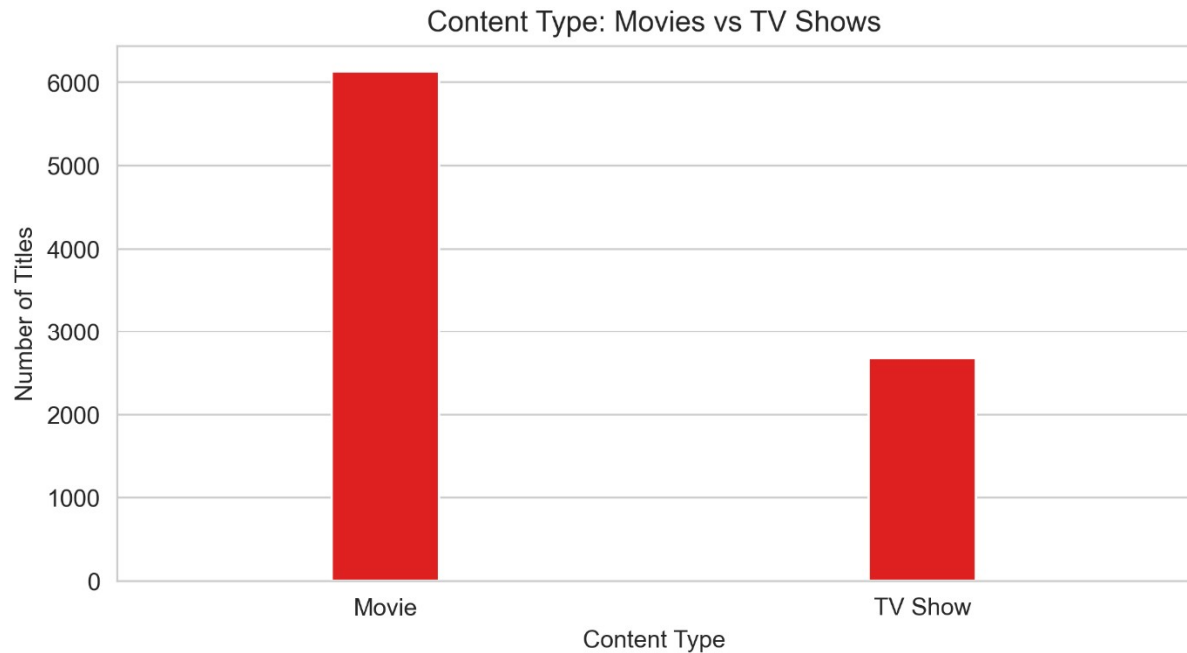
Pre-processing note: 'cast', 'director', 'country' and 'listed_in' are multi-value fields (comma-separated). These were unnested so each row represents one person/country/genre per title, giving an accurate count instead of treating combinations as unique strings.

Fixed a data inconsistency by moving movie duration values from the rating column to the duration column for three records and setting the incorrect rating values to missing.

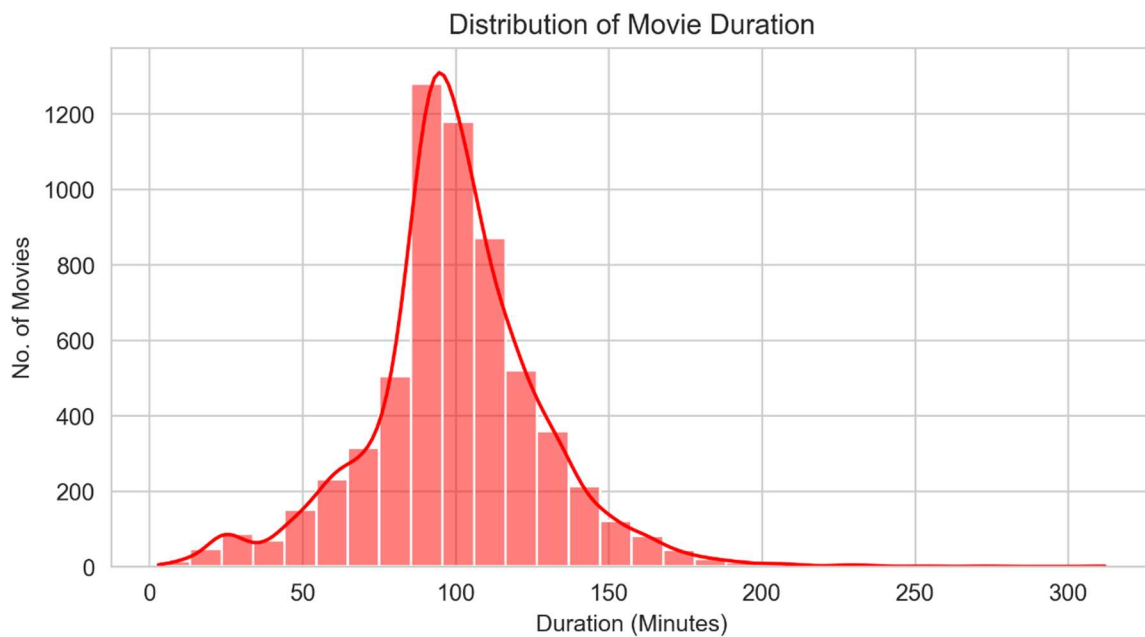
4.1 Univariate Analysis



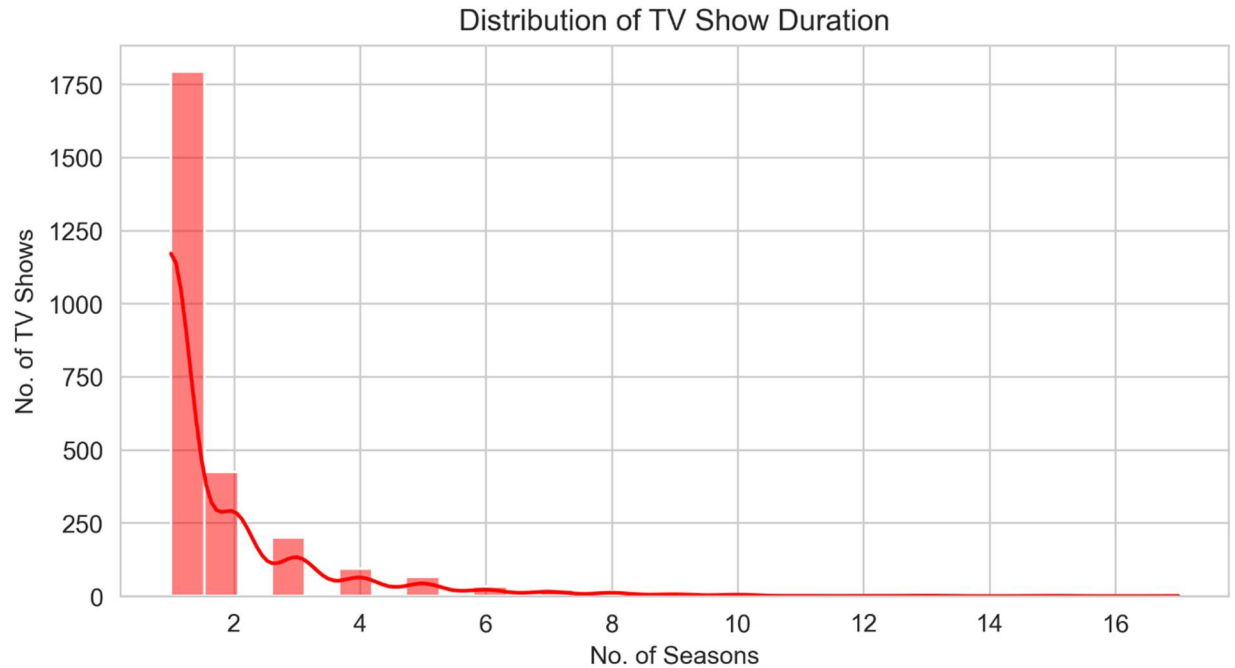
Observation: The release year distribution is strongly right-skewed. Very few titles were released before 2000, showing that Netflix's catalog mainly consists of recent content rather than older classics.



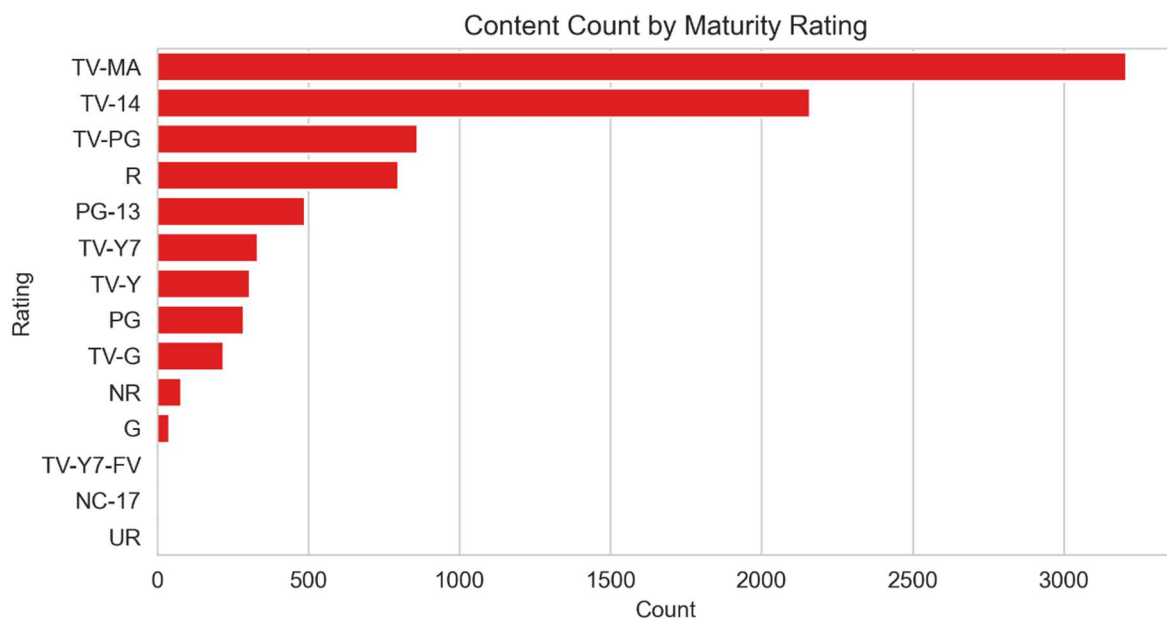
Observation: Movies are more common than TV Shows on Netflix. There are 6131 Movies and 2676 TV Shows.



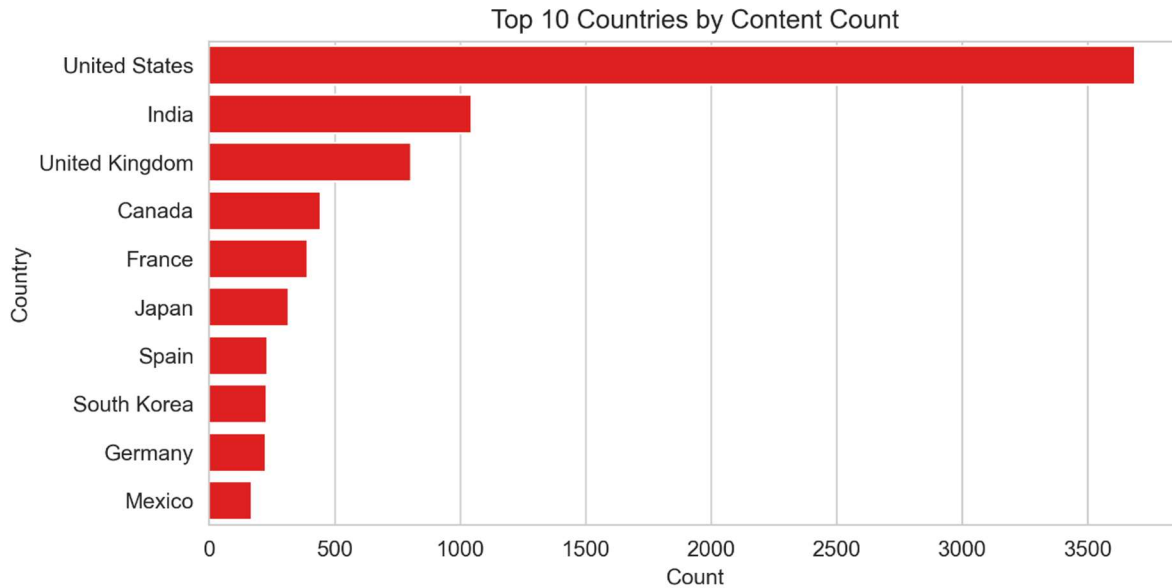
Observation: Movies are usually around 100 minutes long. Most movies are between 87 and 114 minutes, with very few shorter than 40 minutes or longer than 180 minutes. This shows that most movies have a standard length.



Observation: Most TV Shows have only 1 season. About 67% of TV Shows were not renewed beyond the first season, while only a few have many seasons, with the longest running up to 17 seasons.

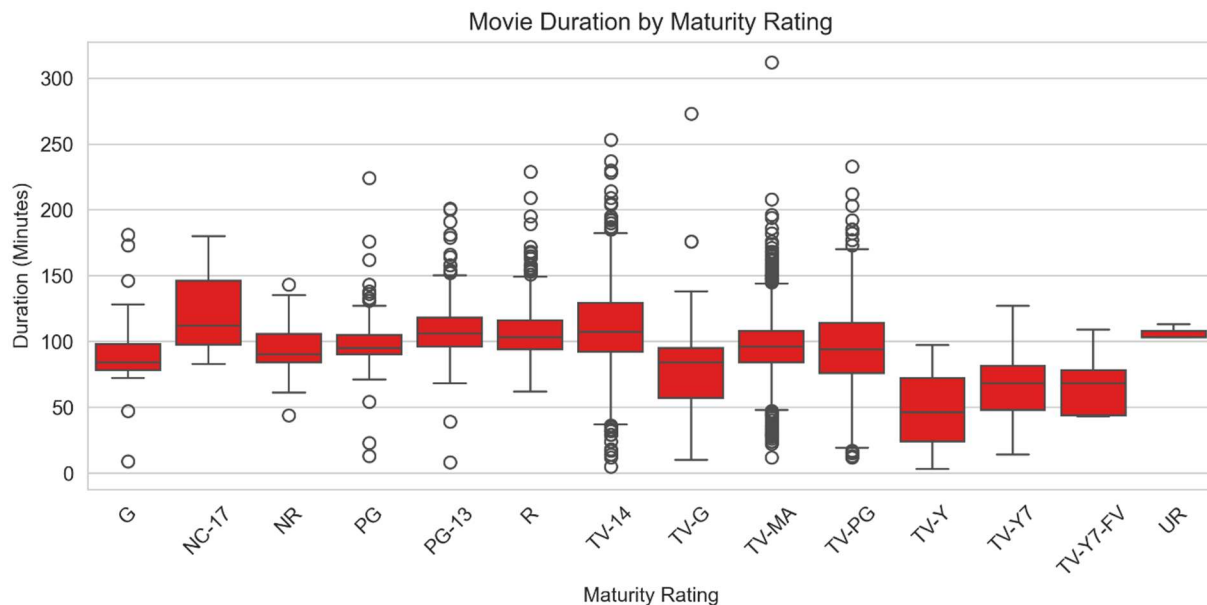


Observation: The most common ratings on Netflix are TV-MA and TV-14, making up about 61% of the catalog. This shows that Netflix mainly focuses on content for teens and adults, while content for young children is much less common.

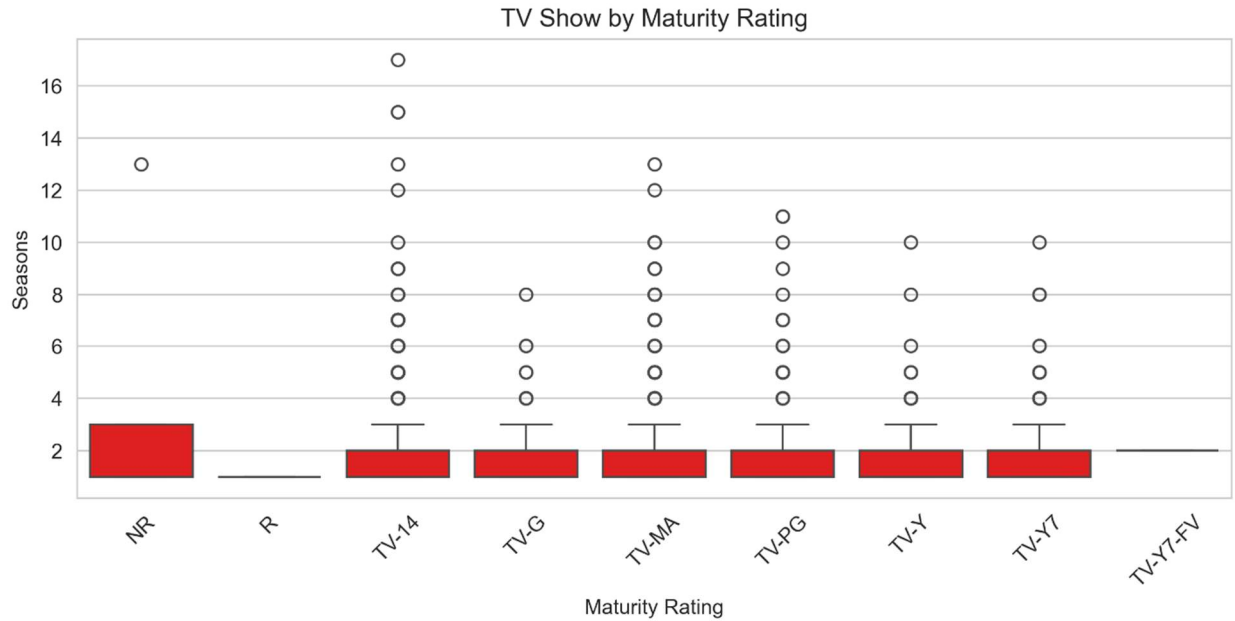


Observation: The United States has the highest number of Netflix titles, followed by India, United Kingdom, Canada, and France. This shows that most Netflix content comes from a few countries, while many other countries have much less content.

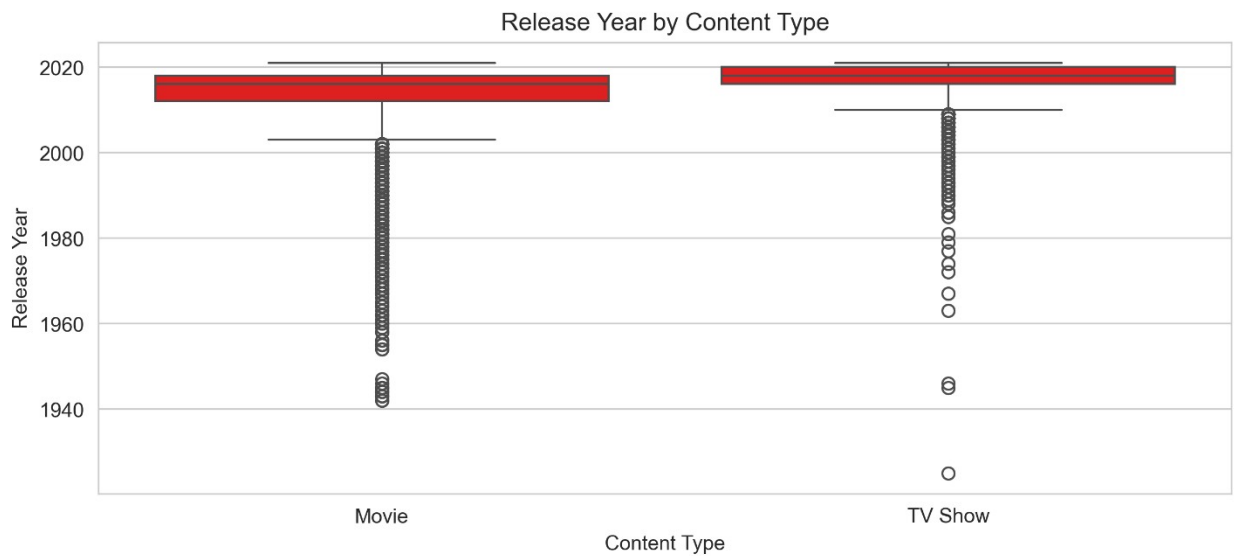
4.2 Bivariate Analysis



Observation: Movies with NC-17 and TV-14 ratings have the longest median runtime, while TV-PG and PG movies are generally shorter. A few movies in several rating categories are over 250 minutes long, showing the presence of some very long films.



Observation: Most TV Shows have a median of 1 season across all rating categories. However, TV-MA and TV-14 have the greatest variation, with some shows running for 10 or more seasons, showing that long-running series are mostly in these ratings.



Observation: TV Shows have a later median release year (2018) than Movies (2016). This shows that Netflix has added more recent TV Shows, while Movies include more older releases.

5. Missing Value & Outlier Check

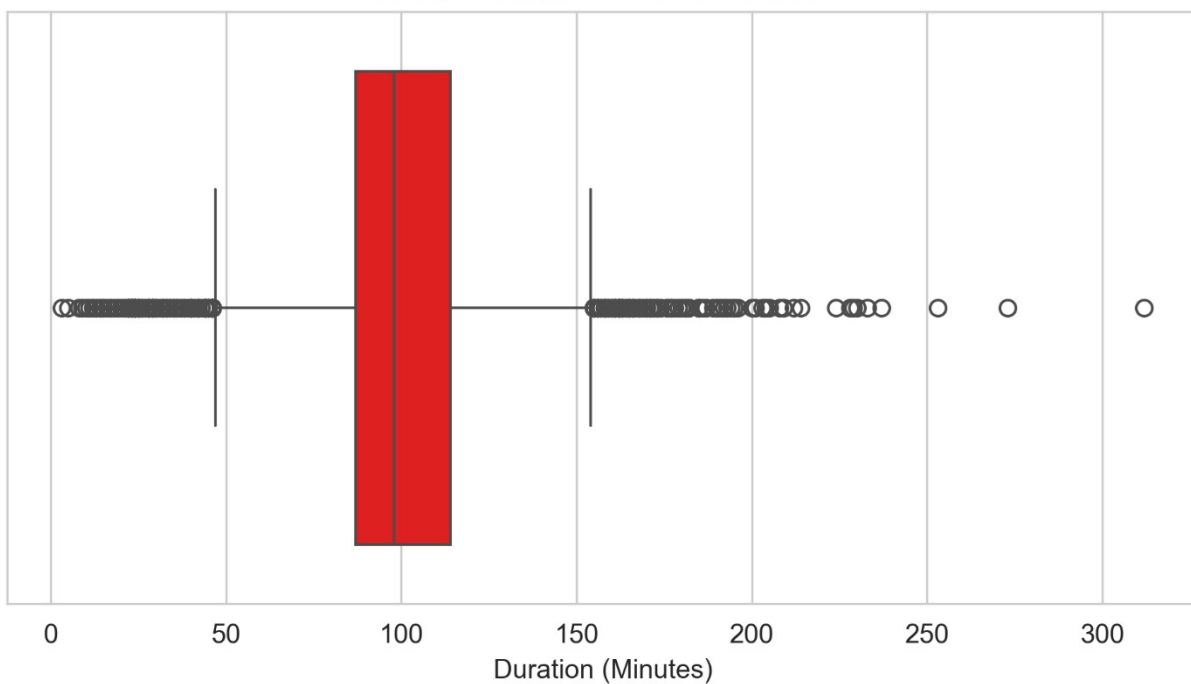
Missing Values

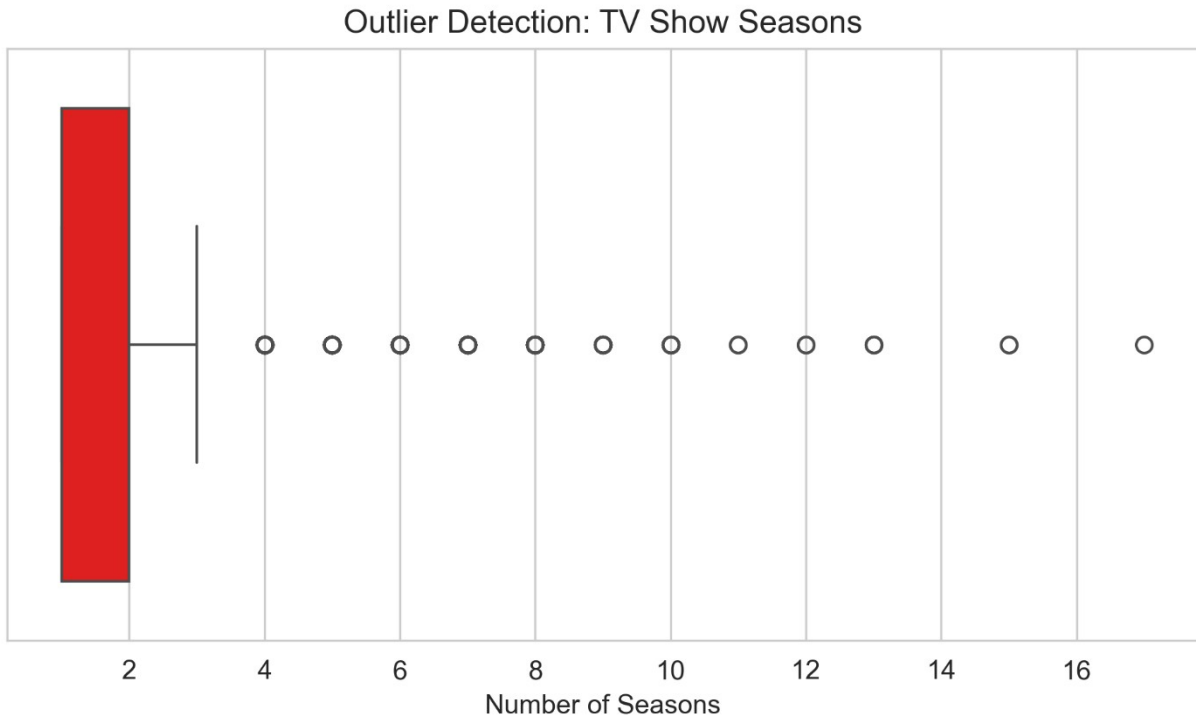
Column	Missing Count	Missing %
director	2,634	29.91%
cast	825	9.37%
country	831	9.44%
date_added	10	0.11%
rating	4	0.05%
duration	3	0.03%

Observations:

- Missing values are mainly present in the director, cast, country, date_added, and rating columns.
- The missing values in the rating column were corrected during data preprocessing, while the remaining missing values were retained as they do not significantly impact the current analysis.

Outlier Detection: Movie Duration





Observation:

- The movie duration boxplot shows a few high-duration outliers, indicating some movies are significantly longer than the typical runtime.
- The TV show seasons boxplot also contains outliers, representing a small number of long-running series with many seasons. These outliers are valid observations and were not removed

6. Insights Based on Non-Graphical and Visual Analysis

6.1 Comments on the Range of Attributes

- Release Year: Ranges from 1925 to 2021 (96 years wide), but 75% of content is from 2013 or later, the effective working range is much narrower than the raw min–max suggests.
- Movie Duration: Ranges from 3 to 312 minutes. The bulk sits between 87–114 minutes (IQR), extremes are rare specials/epics rather than the norm.
- TV Show Seasons: Ranges from 1 to 17 seasons, but is extremely narrow in practice where 75% of shows have 2 seasons or fewer.
- Date: Spans 2008 to 2021, but real catalog growth only becomes visible from around 2015 onward, pre-2015 additions are negligible.
- Rating: Has 17 distinct categories, but is effectively concentrated in just 4-5 dominant ratings- TV-MA, TV-14, TV-PG, R, PG-13.
- Country: Spans 86+ distinct countries, but is heavily concentrated, the top 2 countries (US, India) account for a disproportionate share of total volume.

6.2 Distribution & Relationships Between Variables

- Release Year: Most Netflix titles were released between 2017 and 2019, showing that the platform mainly focuses on recent content rather than older titles.
- Movie Duration: Most movies are around 98 minutes long, showing that Netflix movies generally follow a standard movie runtime.
- TV Show Seasons: Most TV Shows have only 1 season. TV-MA and TV-14 shows have the widest range of seasons, including many long-running series.
- Type vs Country: Content preferences vary by country. South Korea and Japan have more TV Shows, while India and Germany have more Movies, showing that content type differs across countries.

6.3 Comments for Each Univariate and Bivariate Plot

- **Distribution of Release Year (Histogram):** Most Netflix titles were released after 2000, with a rapid increase after 2015. This shows that Netflix mainly focuses on recent content.
- **Content Type: Movies vs TV Shows (Countplot):** Movies are much more common than TV Shows on Netflix. This shows that Netflix's catalog is mainly focused on Movies.

- **Distribution of Movie Duration (Histogram):** Most movies are between 80 and 120 minutes long. Only a few movies have very long runtimes.
- **Distribution of TV Show Seasons (Histogram):** Most TV Shows have only one season. The number of shows decreases as the number of seasons increases, so long-running series are less common.
- **Content Count by Maturity Rating (Countplot):** TV-MA and TV-14 are the most common maturity ratings. This shows that Netflix mainly offers content for teens and adults.
- **Top 10 Countries by Content Count (Bar Plot):** The United States has the highest number of titles, followed by India and the United Kingdom. Most Netflix content comes from a few major countries.
- **Movie Duration by Maturity Rating (Boxplot):** Movies with TV-MA, TV-14, R, and PG-13 ratings are generally longer than children's movies. These ratings also have a wider range of movie durations.
- **TV Show Seasons by Maturity Rating (Boxplot):** Most TV Shows have one season across all maturity ratings. TV-MA and TV-14 include more long-running shows with a higher number of seasons.
- **Release Year by Content Type (Boxplot):** Both Movies and TV Shows are mostly recent releases. However, TV Shows have a slightly more recent release year than Movies, showing that Netflix has added more TV Shows in recent years.

7. Business Insights

- Movies still dominate the catalog, but TV Shows are growing. The share of TV Show additions increased from 25% to 33.7% between 2018 and 2021, while Movie additions declined after 2020.
- Most TV Shows have only one season. About 67% of TV Shows have a single season, showing that long-running series are relatively uncommon.
- Long-running TV Shows are mostly mature-rated. Shows with 4 or more seasons are mainly TV-MA and TV-14, suggesting these categories are more likely to continue for multiple seasons.
- Content preferences vary by country. South Korea and Japan have more TV Shows, while India and Germany have more Movies. This shows that content strategy should be tailored for each country.
- Netflix's catalog is concentrated in a few countries. The United States has the largest number of titles, followed by India. Content from many other countries is available in much smaller numbers.

- Drama, Comedy, and International content are the most common genres. These genres make up a large share of Netflix's catalog, showing they are key content categories. Content additions follow a seasonal pattern. The highest number of titles are added in July and December, while February has the fewest additions. This can help plan future content releases.

8. Recommendations

- Increase investment in TV Shows as their popularity is growing and they help keep viewers engaged for longer.
- Expand family-friendly content to attract more children and families, as the current catalog is mainly focused on teen and adult audiences.
- Continue investing in recent content since most Netflix titles and viewer interest are concentrated on newer releases.
- Develop country-specific content strategies. Increase TV Shows in markets like South Korea and Japan, while continuing strong Movie investments in India and Germany.
- Maintain a balanced mix of Movies and TV Shows to meet different audience preferences and attract a wider subscriber base.
- Invest more in popular genres such as Drama, Comedy, and International content, as they make up a large part of the catalog.
- Schedule major content releases in July and December, when Netflix adds the highest number of titles and audience activity is likely to be higher.
- Work with successful directors, actors, and production studios while also supporting new creators to keep the catalog fresh and diverse.
- Focus on high-quality limited series, as most TV Shows have only one season and can attract viewers with lower long-term production costs.
- Increase content from underrepresented countries to improve regional diversity, reach new audiences, and strengthen Netflix's global presence.